Data Analytics Interim Project Proposal
GROUP 3

Overview

For this project, our team intends to take a structured, step-by-step approach to analyse the Adventure Works demo database and extract meaningful insights based on the questions provided. Our approach will combine SQL, Python, and data analytics techniques to ensure that the analysis is both accurate and actionable.

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Objectives and steps

Objectives

The goal of this project was to answer six different business questions using the AdventureWorks database. Each question focused on a specific part of the business, such as sales, employees, or stores. Our aim was to extract the right data, analyse it correctly, and present clear insights using SQL and Python.

Steps

1. **Understand each question**
   We began by identifying what each question was asking, and which tables and columns were needed to answer it.

2. **Set consistent assumptions**
   We agreed on the assumptions to use across all questions, such as the rules for currency conversion and which PersonTypes count as real employees.

3. **Write the SQL queries**
   For each question, we wrote the SQL needed to extract, clean, and aggregate the relevant data.

4. **Export SQL results**
   The outputs were exported into CSV files so they could be used in Python for further analysis.

5. **Analyse and visualise in Python**
   We loaded the CSV files, inspected the data, and created visualisations (bar charts, scatter plots, log scales, etc.) depending on what was most appropriate for each question.

6. **Interpret the results**
   We analysed what the numbers and graphs meant in practical terms—for example, identifying the best-performing country or checking whether annual leave affects bonuses.

7. **Write conclusions**
   For every question, we wrote a clear conclusion that summarised the main insight and answered the business problem.

8. **Review and correct**
   We checked that the SQL, Python code, and conclusions matched and made corrections where needed—for example, using averages instead of totals when comparing job titles.

Q1. What are the regional sales in the best performing country?

Objective:

- Identify the top-performing country based on relevant performance metrics.
- For countries with multiple regions, determine the highest-performing region within that country.

Required SQL tables and columns:

Sales.SalesTerritory : TerritoryID, Name, CountryRegionCode
Sales.SalesOrderHeader : TotalDue, TerritoryID
Person.CountryRegion : Name, CountryRegionCode

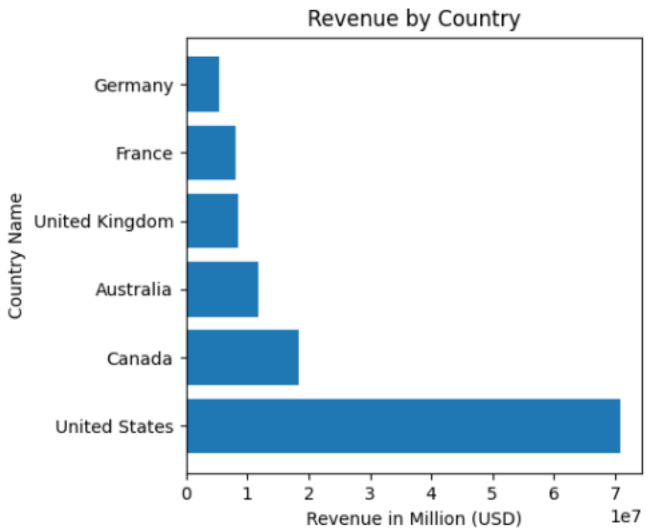
SQL Code:

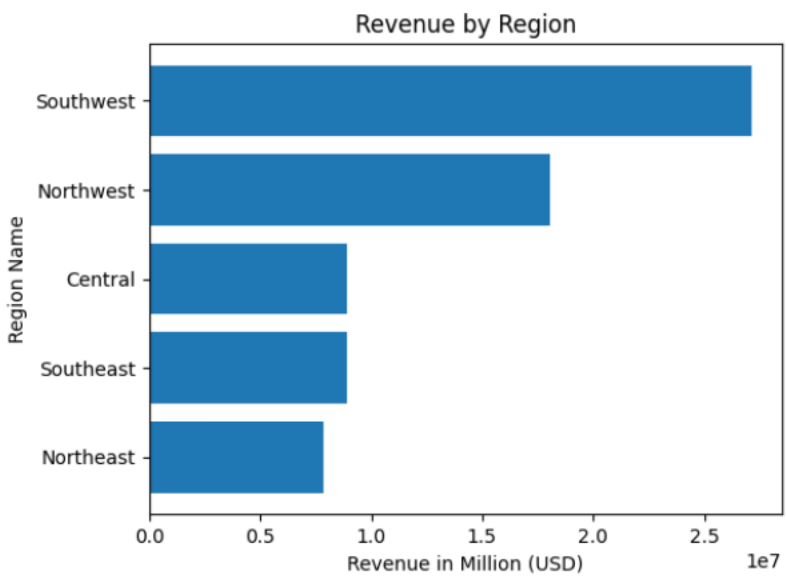
```
SELECT
  st.CountryRegionCode,
  cr.Name AS CountryName,
  SUM(soh.TotalDue) AS Revenue
FROM Sales.SalesOrderHeader AS soh
JOIN Sales.SalesTerritory AS st
  ON soh.TerritoryID = st.TerritoryID
JOIN Person.CountryRegion AS cr
  ON st.CountryRegionCode = cr.CountryRegionCode
GROUP BY
  st.CountryRegionCode,
  cr.Name
ORDER BY
  Revenue DESC;

SELECT
  st.Name AS RegionName,
  SUM(soh.TotalDue) AS Revenue
FROM Sales.SalesOrderHeader AS soh
JOIN Sales.SalesTerritory AS st
  ON soh.TerritoryID = st.TerritoryID
WHERE st.CountryRegionCode = 'US'   -- Because US is classified in regions
GROUP BY st.Name
ORDER BY Revenue DESC;
```

Python Code:

```
import pandas as pd
import matplotlib.pyplot as plt

region = pd.read_csv('python/Q1region.csv', header = None, names = ['Name','Revenue'])
```

```
 # adding header
print(region)

plt.barh(region['Name'], region['Revenue'])
plt.xlabel('Revenue')
plt.ylabel('Region Name')
plt.title('Revenue by Region')
plt.savefig('python/Q1region.png')
plt.show()
```





Conclusion:

We calculated total revenue by country to identify the top-performing country, the United States. We then analysed revenue by region within the U.S. and identified the highest-contributing region. A horizontal bar chart was used to compare regional revenues visually, as it makes differences between regions easy to see: the Southwest region is the region performing better in the US, and in the whole company.

Q2. What is the relationship between annual leave taken and bonus?

Objective:

To determine whether employees who take more annual leave receive higher or lower bonuses.

Required SQL tables and columns:

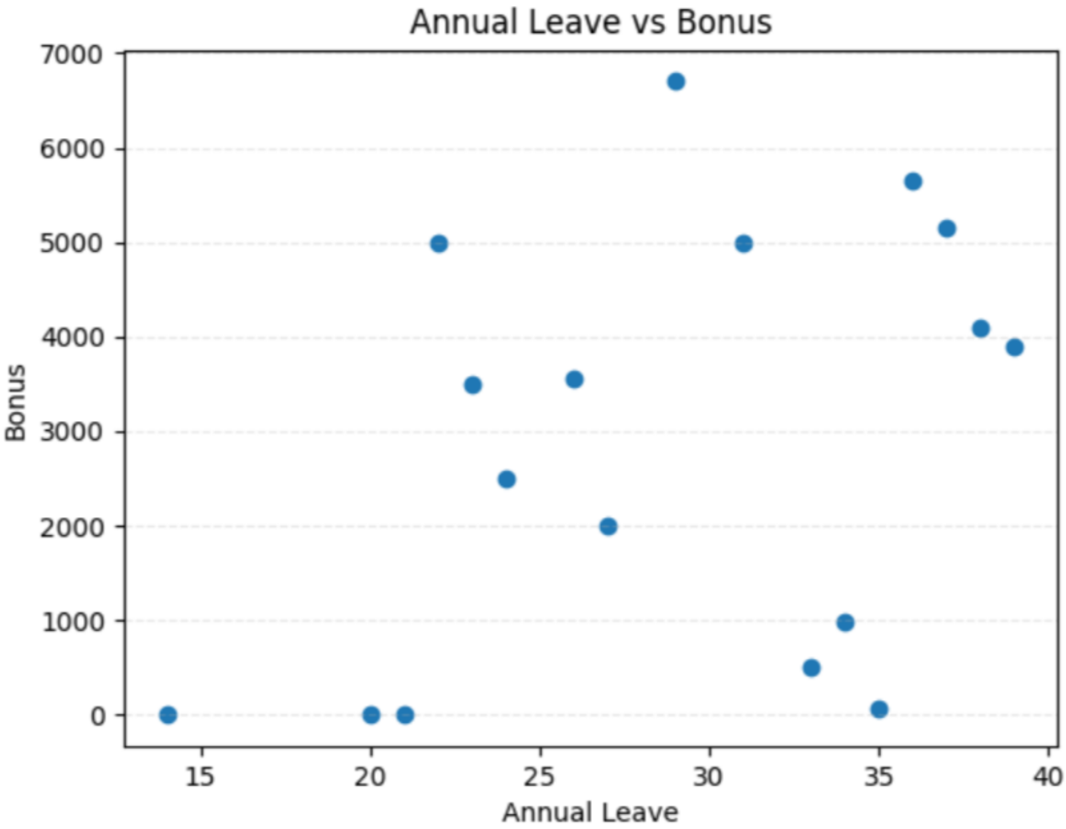
- HumanRessources.Employee : VacationHours = annual leave, BusinessEntityID
- Sales.SalesPerson : Bonus, BusinessEntityID

SQL Code:

```
SELECT em.VacationHours as Annual_Leave, sp.Bonus, em.BusinessEntityID
FROM HumanResources.Employee as em
JOIN Sales.SalesPerson as sp
ON sp.BusinessEntityID = em.BusinessEntityID
ORDER BY em.VacationHours ;
```

Python Code:

```
import pandas as pd
import matplotlib.pyplot as plt

bonus = pd.read_csv('python/Q2bonus.csv', header = None, names = ['Annual
Leave','Bonus', 'ID']) #adding header
print(bonus )
plt.scatter(bonus['Annual Leave'], bonus['Bonus'])
plt.xlabel('Annual Leave')
plt.ylabel('Bonus')
plt.title('Bonus vs Annual Leave')
plt.savefig('python/Q2bonus.png')
plt.show()
Corr = bonus[['Annual Leave', 'Bonus']]
# Print correlation between Annual Leave and Bonus
correlation_value = Corr.corr().loc['Annual Leave', 'Bonus']
print("Correlation between Annual Leave and Bonus:", correlation_value)
```



Conclusion:

The correlation between annual leave and bonus is 0.30, which is very close to zero and indicates a weak and non-meaningful relationship. In practical terms, this means that taking annual leave does not predict whether an employee receives a higher or lower bonus. The scatter plot supports this conclusion: the points are widely scattered with no visible upward or downward trend, showing that bonuses are likely influenced by performance or sales outcomes rather than annual leave taken.

Q3. What is the relationship between Country and Revenue?

Objective:

Determine the total revenue generated by each country.

Required tables and columns:

- Sales.SalesOrderHeader : TotalDue for Revenue, TerritoryID, CurrencyRateID
- Sales.SalesTerritory : TerritoryID, Name, CountryRegionCode
- Person.CountryRegion : Name, CountryRegionCode
- Sales;CurrencyRate : AverageRate, ToCurrencyCode for 'USD'

Assumption:

- If CurrencyRateID is NULL, we assume the order is already in **USD** (company base currency).
- If CurrencyRateID is NOT NULL and ToCurrencyCode = 'USD', we multiply TotalDue by AverageRate to convert it to USD.
- If CurrencyRate doesn't convert to USD, we fall back to TotalDue as-is.

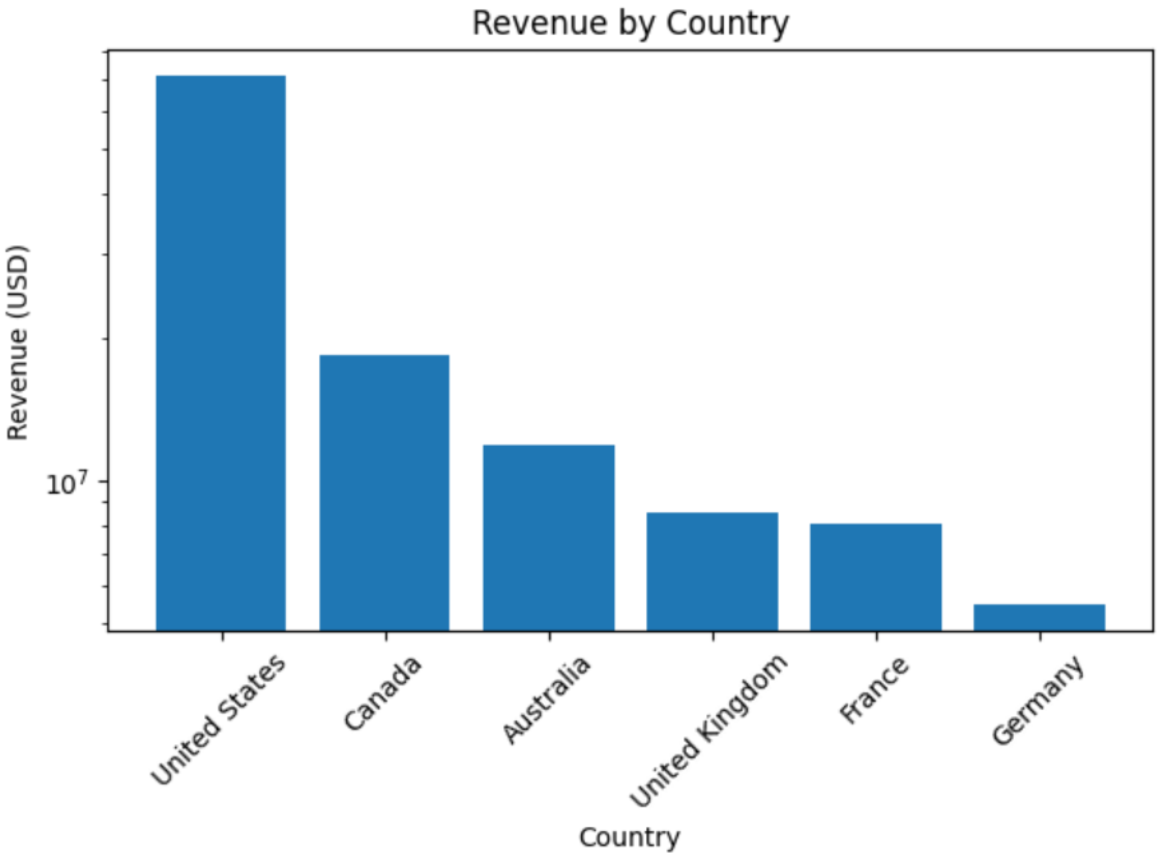
SQL code:

```
SELECT st.CountryRegionCode , cr.Name as Country, SUM (
 CASE
 WHEN soh.CurrencyRateID IS NOT NULL
AND cr2.ToCurrencyCode = 'USD'
 THEN soh.TotalDue * cr2.AverageRate
 ELSE soh.TotalDue
 END) as Revenue
 FROM Sales.SalesOrderHeader as soh
 JOIN Sales.SalesTerritory as st ON st.TerritoryID = soh.TerritoryID
 JOIN Person.CountryRegion as cr ON cr.CountryRegionCode = st.CountryRegionCode
 LEFT JOIN Sales.CurrencyRate as cr2 ON soh.CurrencyRateID = cr2.CurrencyRateID
 GROUP BY st.CountryRegionCode, cr.name
 ORDER BY Revenue DESC;
```

Python code:

```
 import pandas as pd
import matplotlib.pyplot as plt
```

```
CountryRevenue = pd.read_csv('python/Q3RevenueCountry.csv', header = None, names =
['Country','Revenue'])  #adding header
print(CountryRevenue )
plt.bar(CountryRevenue['Country'], CountryRevenue['Revenue'])
plt.yscale("log")  #to change y axis to log scale
plt.xticks(rotation=45)
plt.xlabel('Country')
plt.ylabel('Revenue')
plt.title('Revenue by Country')
plt.tight_layout()
plt.show()
```



Conclusion:

We converted all sales amounts into USD using the CurrencyRate table and then aggregated total
revenue by country to ensure a fair comparison across different markets. This allowed us to identify which countries generate the highest revenue regardless of local currency.
The results show a strong geographical imbalance: the United States generates by far the highest revenue, followed by Canada and Australia, while European countries contribute significantly less. This means that the company’s revenue performance is highly concentrated in a few key markets.

Larger or more active sales territories drive out most of the company’s income, while other countries
play a smaller role in overall revenue. Understanding which countries dominate sales can support better strategic planning, resource allocation, and market development decisions.

Q4. What is the relationship between sick leave and Job Title?

Objective:

Determine whether an employee's Job Title or PersonType has an impact on the amount of sick leave taken.

Required tables and columns:

- HumanRessources.Employee : SickLeaveHours, JobTitle
- Person.Person : BusinessEntityID, PersonType

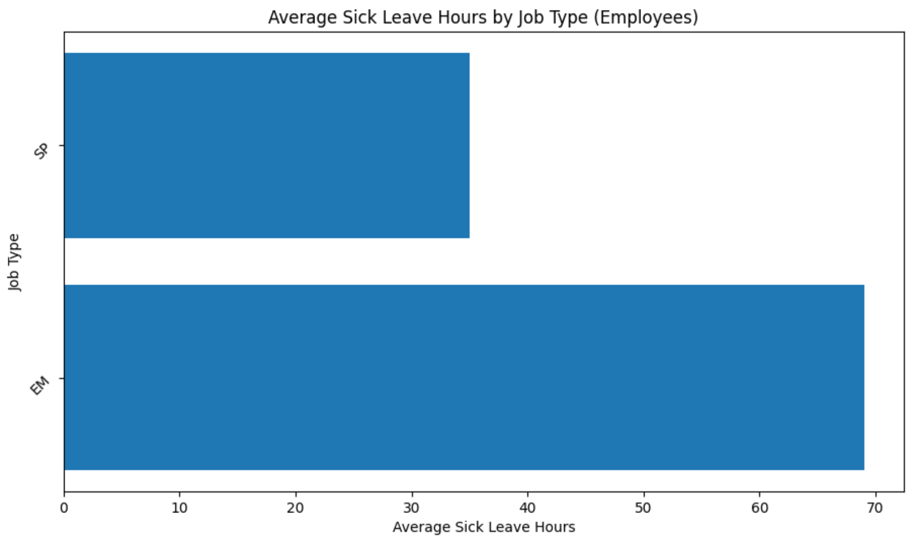
SQL code:

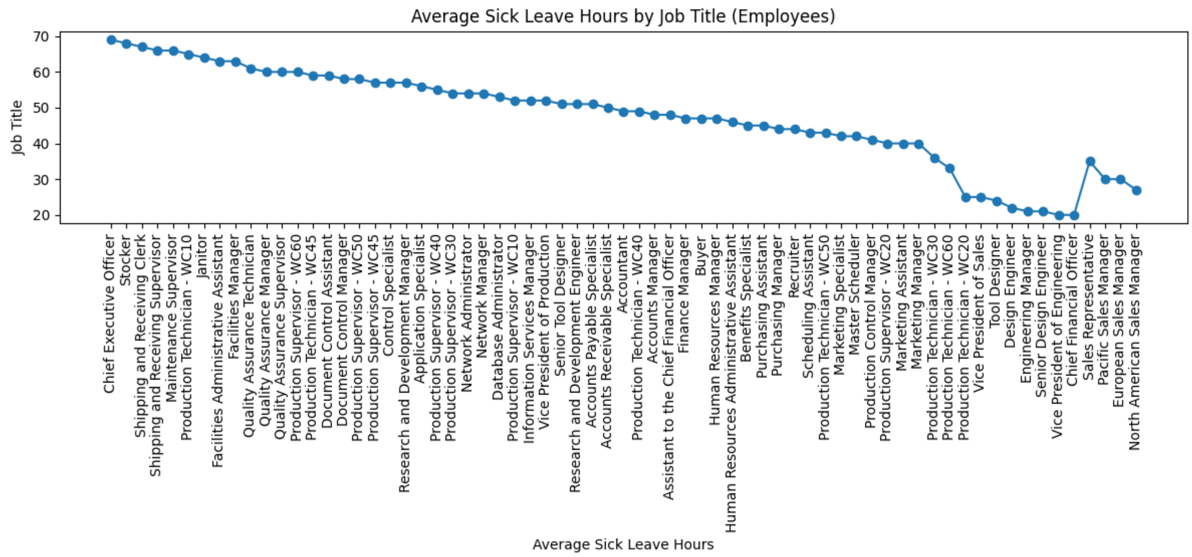
```
SELECT pp.PersonType, em.JobTitle, AVG(em.SickLeaveHours) as avgHours,
SUM(em.SickLeaveHours) as TotalHours, COUNT (*) as NbEmployee
FROM  HumanResources.Employee as em
JOIN Person.Person as pp on pp.BusinessEntityID = em.BusinessEntityID
GROUP BY em.JobTitle, pp.PersonType
ORDER BY avgHours DESC;
```

Python code:

```
import pandas as pd
import matplotlib.pyplot as plt

SickLeave = pd.read_csv(
 'python/Q4sickleave.csv',
 header=None,
names=['Job Title', 'Job Type','Avg Sick Leave', 'Total Sick Leave', 'Num Employees'])

# Plot Figure 1
 plt.figure(figsize=(12, 14))
 plt.barh(SickLeave['Job Type'], SickLeave['Avg Sick Leave'])
 plt.xlabel('Average Sick Leave Hours')
 plt.ylabel('Job Type')
 plt.title('Average Sick Leave Hours by Job Type (Employees)')
 # Rotate y-axis labels for better readability
 plt.yticks(rotation=45)
 plt.tight_layout()
```

```
plt.savefig('python/Q4sickleave1.png')
plt.show()


SickLeave = pd.read_csv(
  'python/Q4sickleave.csv',
 header=None,
 names=['Job Title', 'Job Type','Avg Sick Leave', 'Total Sick Leave', 'Num Employees'])
plt.figure(figsize=(12, 14))
plt.plot(SickLeave['Job Title'], SickLeave['Avg Sick Leave'], marker='o')
plt.xlabel('Average Sick Leave Hours')
plt.ylabel('Job Title')
plt.title('Average Sick Leave Hours by Job Title (Employees)')
# Rotate y-axis labels for better readability
plt.xticks(rotation=90)
plt.tight_layout()

plt.show()
```





Conclusion:

We found that only individuals with PersonType ‘EM’ (Employee) and ‘SP’ (Salesperson) appear in the HumanResources.Employee table, meaning they are the only ones with real job titles and recorded sick-leave hours. All other PersonTypes do not represent internal staff, which is why they were excluded.

After analysing both EM and SP roles, we observed clear differences in average sick-leave hours across job titles. Some positions consistently take more sick leave than others, showing that job role does influence sick-leave behaviour. This suggests that differences may be linked to job demands, workload, or working conditions associated with specific roles.

Q5. What is the relationship between store trading duration and revenue?

Objective:

Determine whether the store's trading duration (in years) has an impact on its revenue.

Required tables and columns:

- Sales.vStoreWithDemographics : YearOpened, StoreName
- Sales.Customer : to find link between store and customers
- Sales.SalesOrderHeader : revenue
- Sales.CurrencyRate : conversion to USD (following Q3)

SQL code:

Code with no currency conversion:

```
SELECT
 vsd.BusinessEntityID AS StoreID,
 vsd.Name AS StoreName,
 vsd.YearOpened,
 YEAR(GETDATE()) - vsd.YearOpened AS TradingDurationYears,
 SUM(soh.TotalDue) AS Revenue
 FROM Sales.vStoreWithDemographics AS vsd
JOIN Sales.Customer AS sc ON sc.StoreID = vsd.BusinessEntityID
JOIN Sales.SalesOrderHeader AS soh ON soh.CustomerID = sc.CustomerID
GROUP BY vsd.BusinessEntityID, vsd.Name, vsd.YearOpened;
```

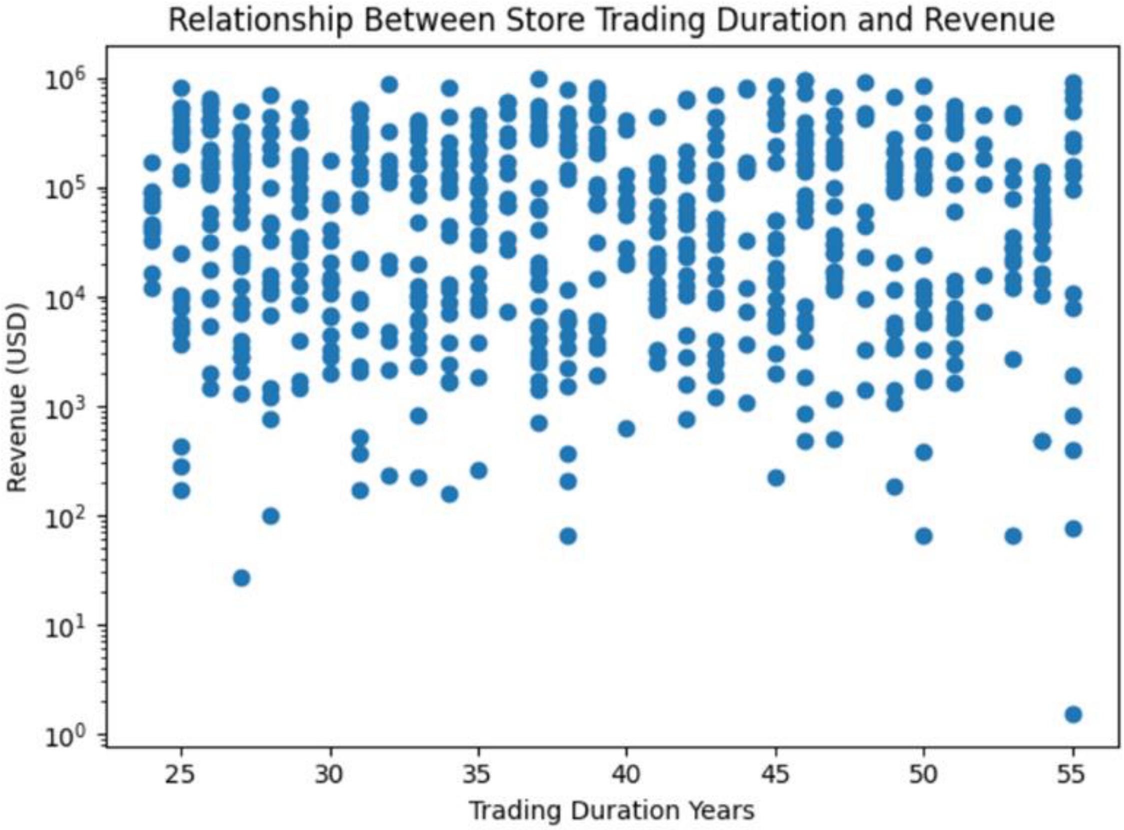
This matches the same assumption as Q3 for currency conversion:

```
SELECT
   vsd.BusinessEntityID AS StoreID,
   vsd.Name AS StoreName,
   vsd.YearOpened,
   YEAR(GETDATE()) - vsd.YearOpened AS TradingDurationYears,
   SUM(
      CASE
         WHEN soh.CurrencyRateID IS NULL THEN soh.TotalDue
```

```
WHEN cr2.ToCurrencyCode = 'USD' THEN soh.TotalDue * cr2.AverageRate
ELSE soh.TotalDue
END
) AS RevenueUSD
FROM Sales.vStoreWithDemographics AS vsd
JOIN Sales.Customer AS sc ON sc.StoreID = vsd.BusinessEntityID
JOIN Sales.SalesOrderHeader AS soh ON soh.CustomerID = sc.CustomerID
LEFT JOIN Sales.CurrencyRate AS cr2 ON soh.CurrencyRateID = cr2.CurrencyRateID
GROUP BY vsd.BusinessEntityID, vsd.Name, vsd.YearOpened
HAVING SUM(
CASE
WHEN soh.CurrencyRateID IS NULL THEN soh.TotalDue
WHEN cr2.ToCurrencyCode = 'USD' THEN soh.TotalDue * cr2.AverageRate
ELSE soh.TotalDue
END
) > 0
ORDER BY vsd.BusinessEntityID DESC;
```

Python code:

```
import pandas as pd
import matplotlib.pyplot as plt
DurationRevenue = pd.read_csv('python/Q5durationRevenue.csv', header = None, names=
['Store ID','Store Name','Year Opened','Trading Duration Years','Revenue'])#adding header
print(DurationRevenue )

plt.scatter(DurationRevenue['Trading Duration Years'], DurationRevenue['Revenue'])
plt.xlabel('Trading Duration Years')
plt.ylabel('Revenue (USD)')
plt.title('Relationship Between Store Trading Duration and Revenue')
Plt.yscale(‘log’)
Plt.tight.layout()
plt.savefig('python/Q5durationRevenue.png')
plt.show()
```



Conclusion:

We present two versions of the query: one without currency conversion and one that converts all revenue into USD using the same assumptions as Q3. The first query shows raw revenue, while the second ensures all stores are compared consistently across currencies. For analysis, we use the converted version (RevenueUSD) because it provides a fair comparison between stores.

We analysed store revenue using consistent USD conversion and compared it with trading duration. The results show no significant correlation, meaning a store's age does not predict its revenue.

Q6. What is the relationship between the size of the stores, number of employees and revenue?

Objective:

To ddetermine whether store size has an impact on the number of employees and on overall store revenue.

SQL table and columns:

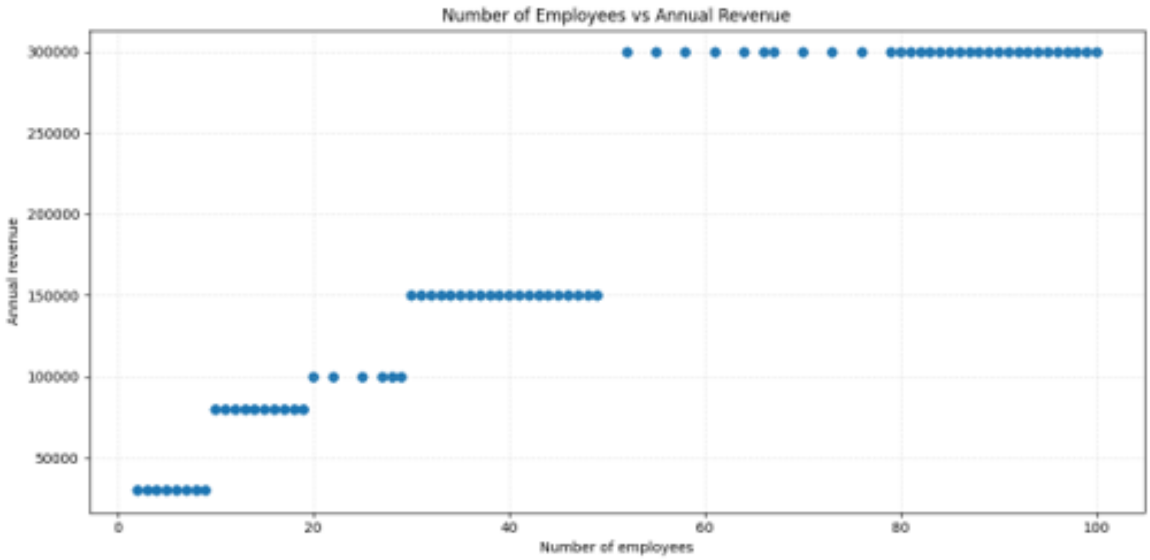
Sales.vStoreWithDemographics : NumberEmployees, SquareFeet, AnnualRevenue

SQL code

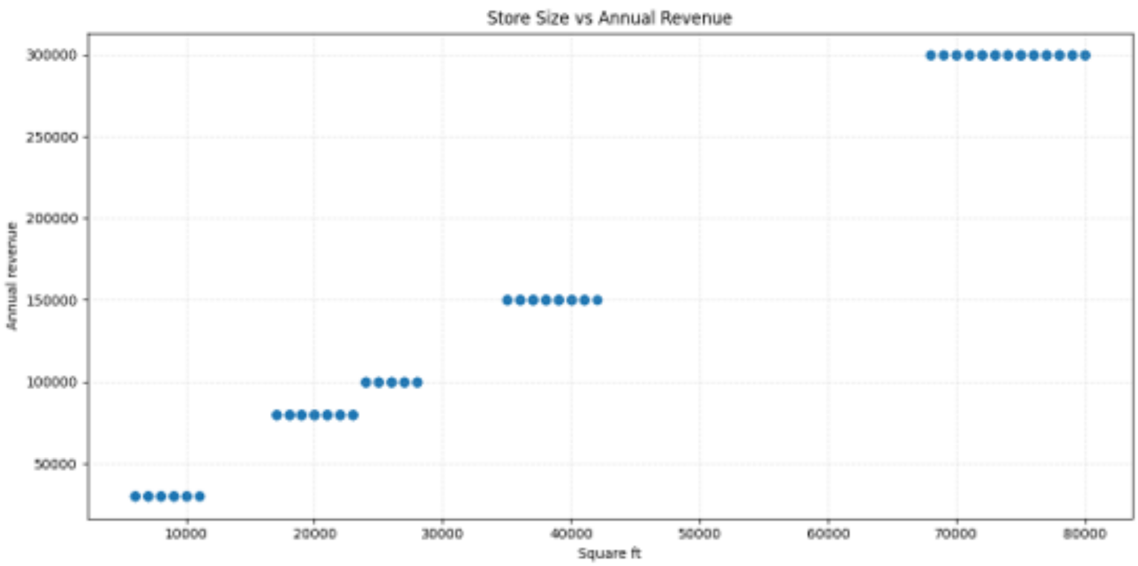
```
SELECT NumberEmployees,
 SquareFeet, AnnualRevenue,
 FROM Sales.vStoreWithDemographics
 ORDER BY NumberEmployees;
```

Python code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\amira\OneDrive\Documents\PYTHON\Group-Project\Q6.csv",
header=None, names=['Number of employees','Square ft','Annual revenue','Number stores'])
print(df.columns)

plt.scatter(df["Number of employees"], df["Annual revenue"])
plt.xlabel("Number of employees")
plt.ylabel("Annual revenue")
plt.title("Number of Employees vs Annual Revenue")
plt.tight_layout()
plt.grid(True, linestyle='--', alpha=0.3)
plt.show()
```



This graph compares the number of employees in each store with the store's annual revenue.
Even without calculating a correlation, the trend is clear: as the number of employees increases, revenue also increases.
Small stores with fewer than 10 employees generate low revenue, while stores with 70–100 employees consistently appear in the highest revenue bracket.
This suggests that staffing levels and revenue grow together, meaning larger teams support higher store performance.



```
plt.scatter(df["Square ft"], df["Annual revenue"])
plt.xlabel("Square ft")
plt.ylabel("Annual revenue")
plt.title("Store Size vs Annual Revenue")
plt.tight_layout()
plt.grid(True, linestyle='--', alpha=0.3)
plt.show()
```

This graph compares store size, measured in square feet, with each store's annual revenue. We can clearly see that revenue increases as stores get larger. Small stores around 8,000–20,000 sq ft generate the lowest revenue, mid-sized stores around 30,000–40,000 sq ft sit in the middle range, and the very large stores above 70,000 sq ft consistently appear in the highest revenue group.
This pattern suggests that bigger stores have more capacity — more space, more stock, more customers — and therefore tend to generate higher revenue.


```
plt.scatter(df["Square ft"], df["Number of employees"])
plt.xlabel("Square ft")
plt.ylabel("Number of employees")
plt.title("Store Size vs Number of Employees")
plt.tight_layout()
plt.grid(True, linestyle='--', alpha=0.3)
plt.show()
```

The last graph compares store size with the number of employees.
The pattern is very clear: the bigger the store, the more employees it has.
Small stores under 15,000 sq ft have only a few staff, mid-sized stores between 20,000–40,000 sq ft typically have 15 to 40 employees, and the very large stores above 70,000 sq ft have the highest staffing levels, often between 70 and 100 employees.
This shows that staffing scales with store capacity — larger stores require more people to operate.

Conclusion:

For this question, we wanted to understand whether larger stores employ more people, and whether store size or staffing is linked to higher revenue. To do this, we used the table Sales.vStoreWithDemographics, which contains:

- NumberEmployees
- SquareFeet
- AnnualRevenue

We selected these three variables for all stores, without grouping, because we needed to compare each store individually. Then, we plotted three scatter graphs in Python to visualise the relationships:

- Store size vs. number of employees
- Store size vs. annual revenue
- Number of employees vs. annual revenue This allowed us to observe patterns across all 701 stores.

Across all stores, we observed a clear visual pattern:
larger stores consistently have more employees, and both store size and staffing are associated with higher revenue.
Small stores cluster in the low-size, low-staff, low-revenue areas, while the biggest stores are found in the high-revenue range.
So overall, store size and number of employees move together with revenue levels.
This doesn't prove a mathematical correlation, but the visual analysis strongly suggests that store size and staffing are key factors linked to store performance.